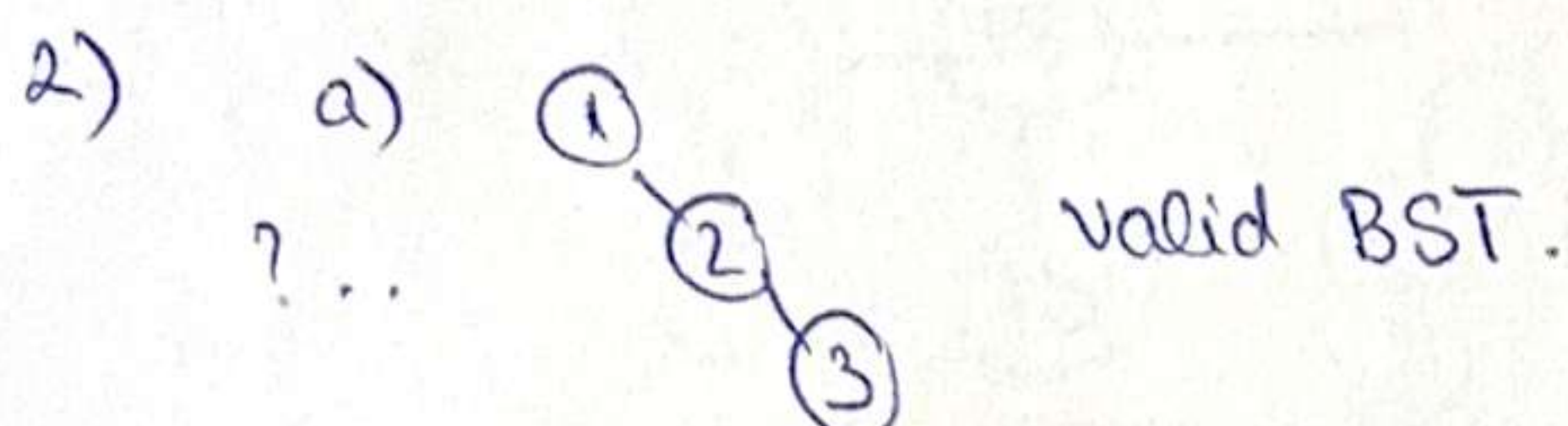


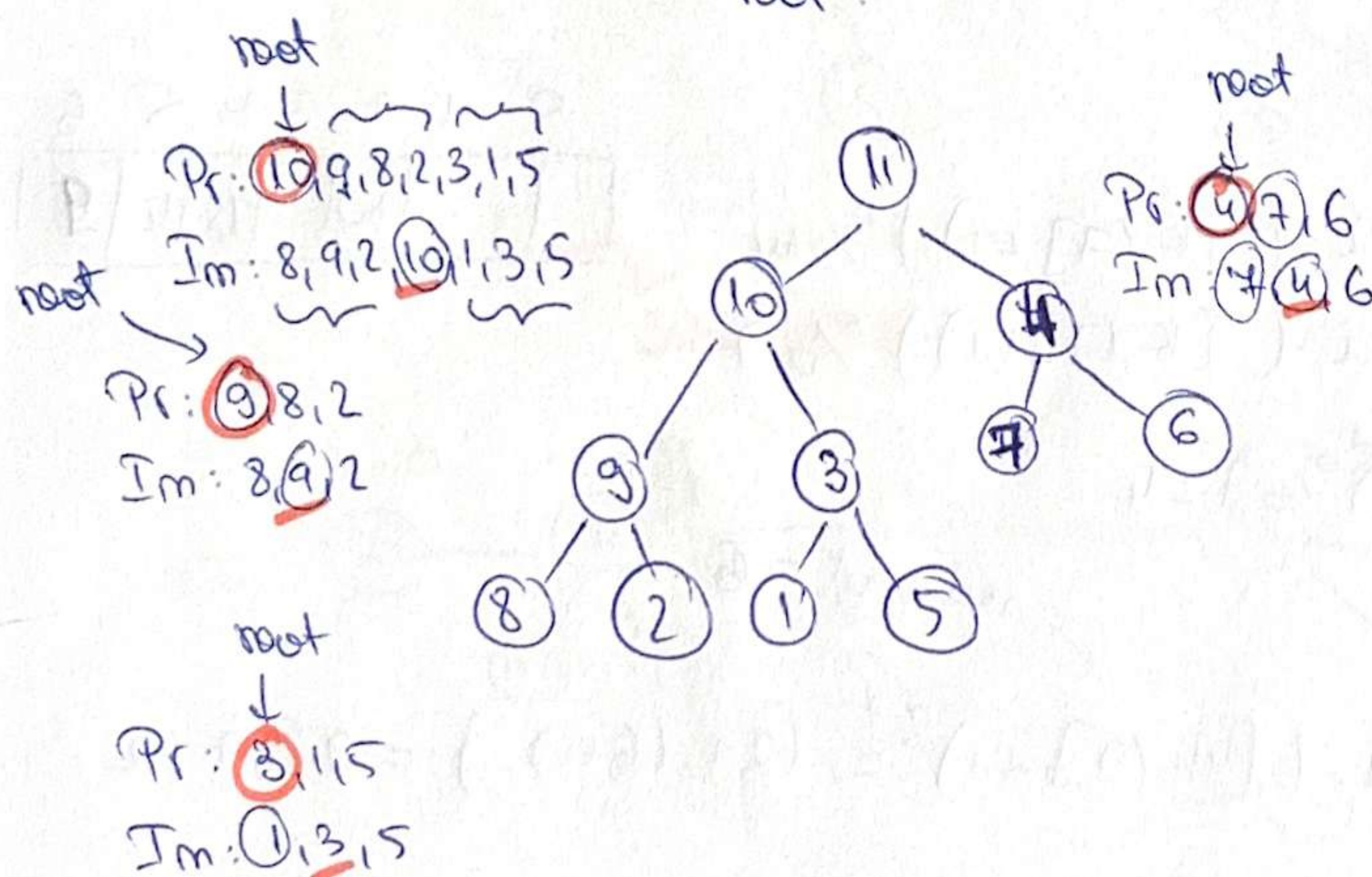
I. 1) $A(n) = 15n^3 + 1089n^2 + 5n \cdot \log_2 n$

$\lim_{n \rightarrow \infty} \frac{A(n)}{n^2} = \infty \Rightarrow A$ grows faster than $n^2 \Rightarrow A \in \Omega(n^2) \Rightarrow \boxed{b) \Omega(n^2)}$

$\lim_{n \rightarrow \infty} \frac{A(n)}{n^3} = 15 > 0 \Rightarrow A \in \Theta(n^3) \Rightarrow \boxed{e) \Theta(n^3)}$



3) INORDER given: 8, 9, 2, 10, 1, 3, 5, 11, 7, 4, 6.
 PREORDER given: 11, 10, 9, 8, 2, 3, 1, 5, 4, 7, 6



4) Are linear probing and double hashing the same when $h''(k) = 1$?

$h_1(k, i) = (h'(k) + i) \% m$ $h_2(k, i) = (h'(k) + i \cdot h''(k)) \% m$
 if $h''(k) = 1 \Rightarrow h_2(k, i) = h_1(k, i)$

6) $m = 7$, open addr, elems: 18, 30, 44, 12, 9. ; $m = 5$. $\alpha = \frac{m}{n} = \frac{5}{7}$

a) linear probing: $h(k, i) = (k \% m + i) \% m = (k \% 7 + i) \% 7$

$h(18, 0) = (4 + 0) \% 7 = 4 \checkmark$

$h(30, 0) = 2 \checkmark$

$h(44, 0) = 2 \Rightarrow$ collision

$h(44, 1) = 3 \checkmark$

$h(12, 0) = 5 \checkmark$

$h(9, 0) = 2 \Rightarrow$ coll.

$h(9, 1) = 3 \Rightarrow$ coll.

$h(9, 2) = 4 \Rightarrow$ coll.

$h(9, 3) = 5 \Rightarrow$ coll.

$h(9, 4) = 6 \Rightarrow \checkmark$

0	1	2	3	4	5	6
		30	44	18	12	9

$i = \overline{0, 6}$

b) quadratic probing

$$h(k, i) = (k \% m + i + i^2) \% m = (k \% 7 + i + i^2) \% 7$$

$$h(18, 0) = (18 \% 7 + 0) \% 7 = 4$$

$$h(30, 0) = (30 \% 7 + 0) \% 7 = 2$$

$$h(44, 0) = (44 \% 7 + 0) \% 7 = 2$$

$$h(44, 1) = (2 + 1 + 1) \% 7 = 4$$

$$h(44, 2) = (2 + 2 + 4) \% 7 = 1$$

$$h(12, 0) = (12 \% 7 + 0 + 0) \% 7 = 5$$

$$h(9, 0) = (9 \% 7 + 0 + 0) \% 7 = 2$$

$$h(9, 1) = 4$$

$$h(9, 2) = 1$$

$$h(9, 3) = 0$$

0	1	2	3	4	5	6
9	44	30		18	12	

c) double hashing

$$h(k, i) = (k \% m + i \cdot ([k/7] + 1)) \% m$$

$$h(k, i) = (k \% 7 + i \cdot ([k/7] + 1)) \% 7$$

0	1	2	3	4	5	6
		30		18	12	9

$$h(18, 0) = (18 \% 7 + 0) \% 7 = 4$$

$$h(30, 0) = \dots = 2$$

$$h(44, 0) = \dots = 2$$

$$h(44, 1) = (2 + 1 \cdot ([44/7] + 1)) \% 7 = (2 + (6 + 1)) \% 7 = 9 \% 7 = 2$$

$$h(44, 2) = 2$$

$$h(44, 3) = 2$$

... 44 cannot be added! $h(44, i)$ is always 2, which is occupied already! skipping it!

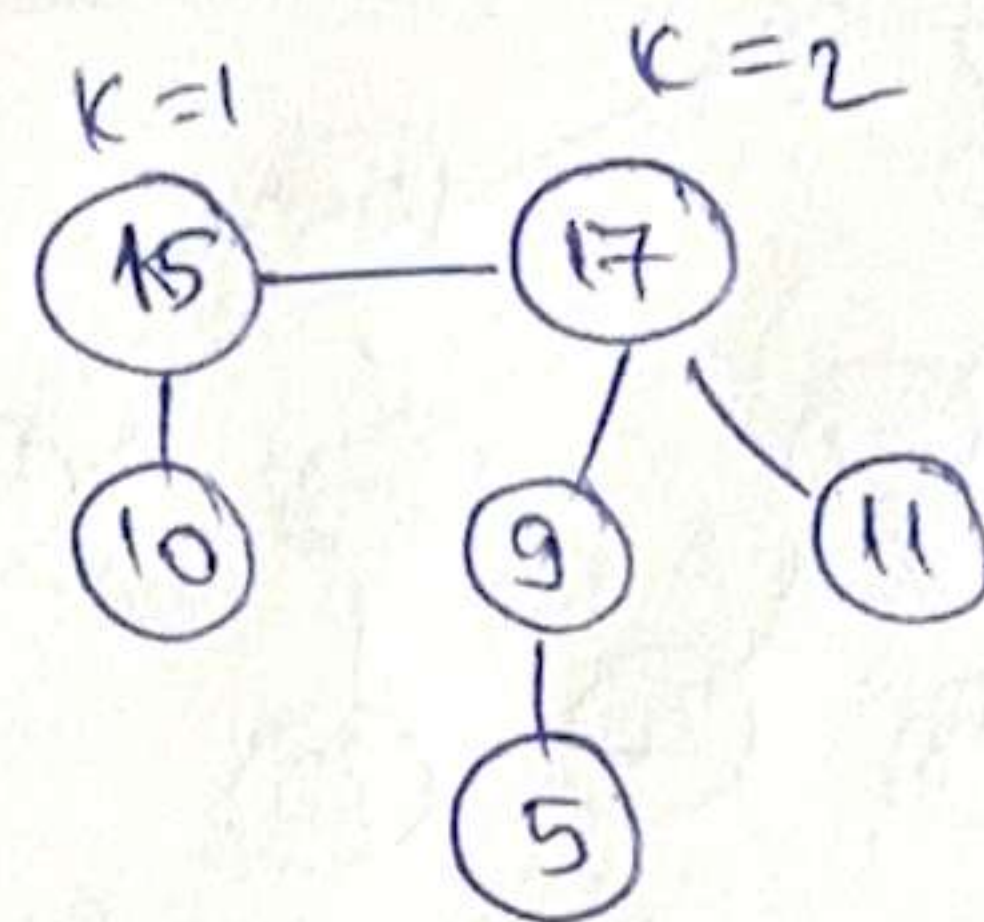
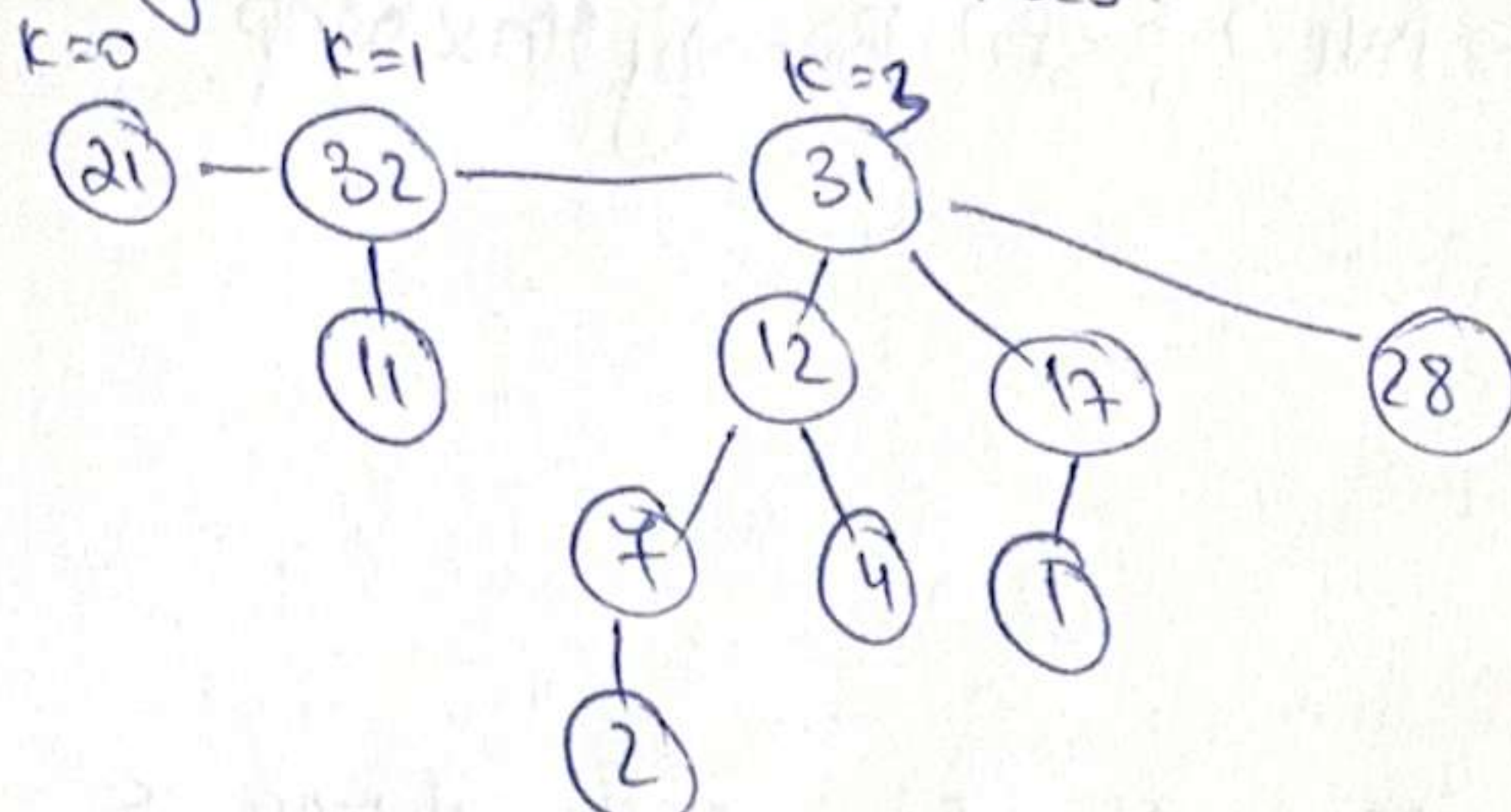
$$h(12, 0) = 5$$

$$h(9, 0) = 2$$

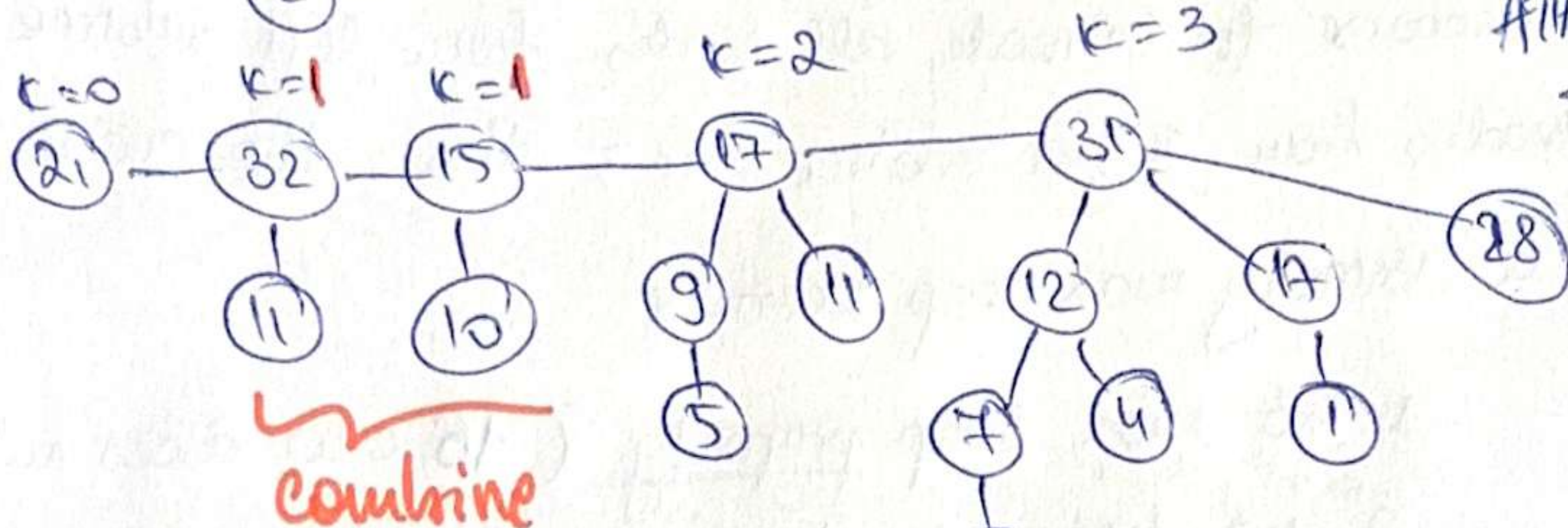
$$h(9, 1) = 4$$

$$h(9, 2) = 6$$

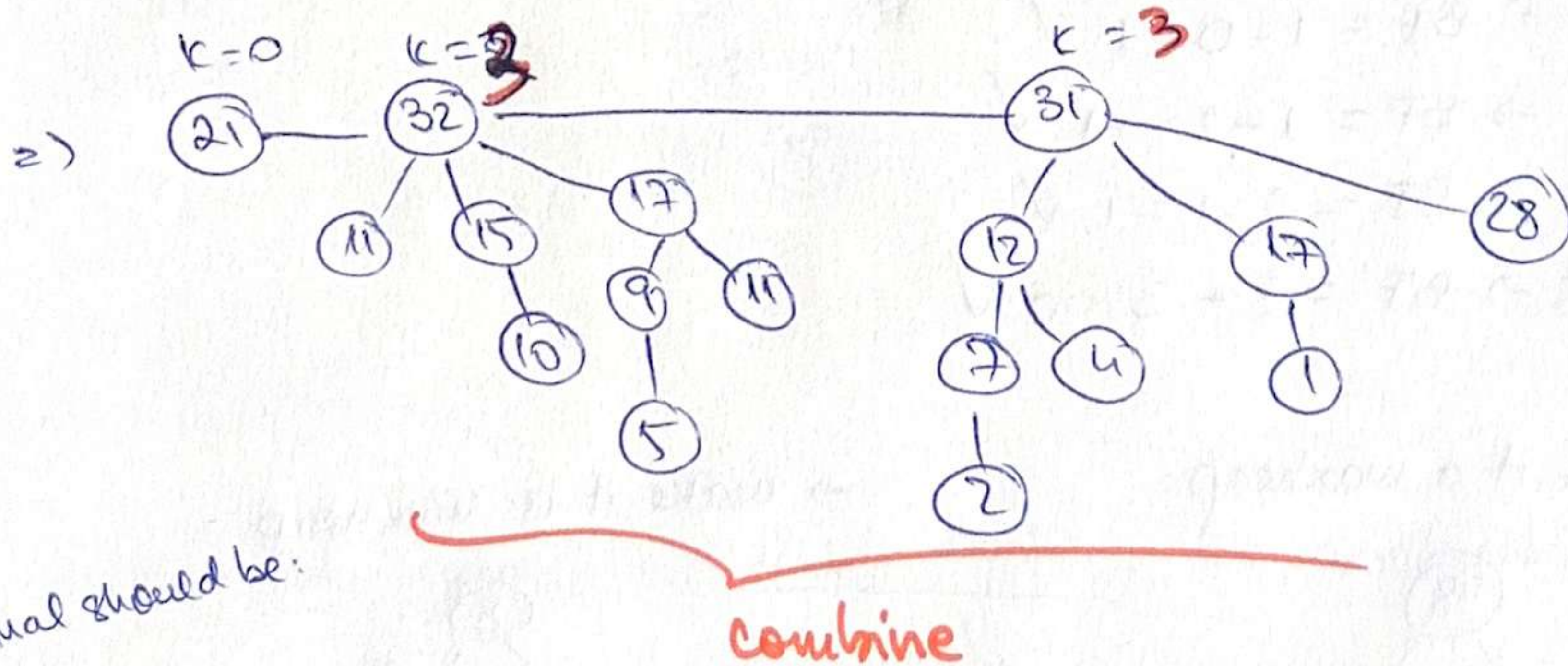
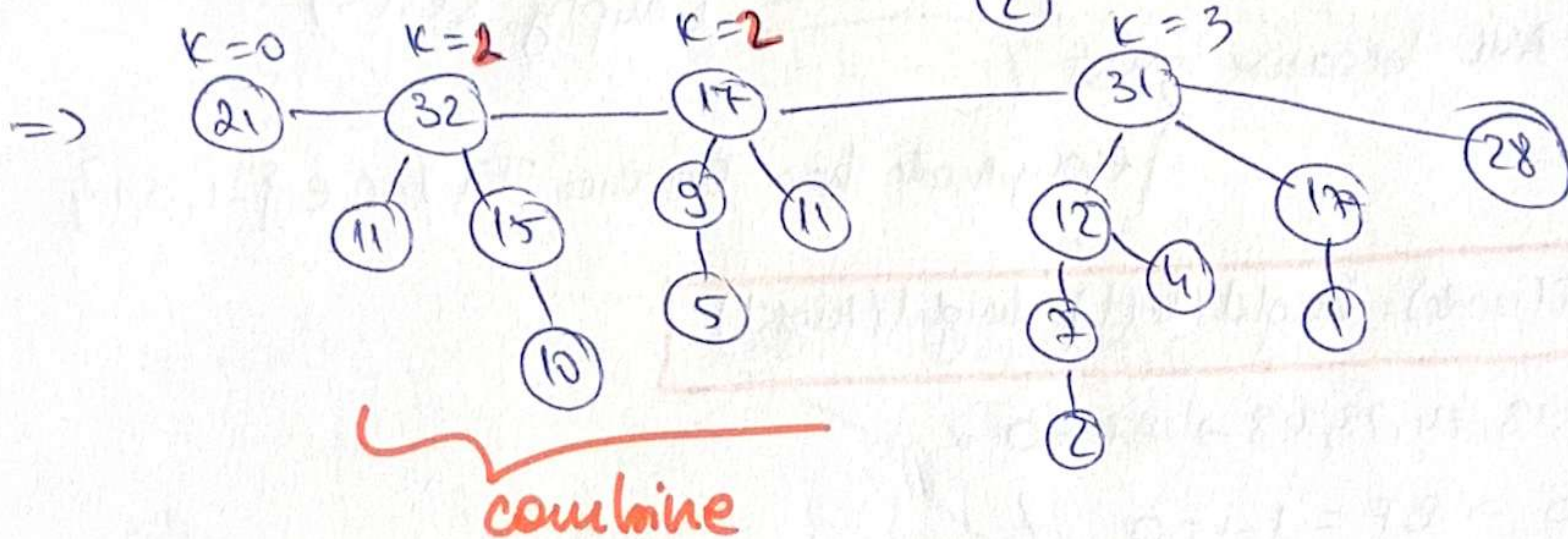
f) Merge the binomial trees:



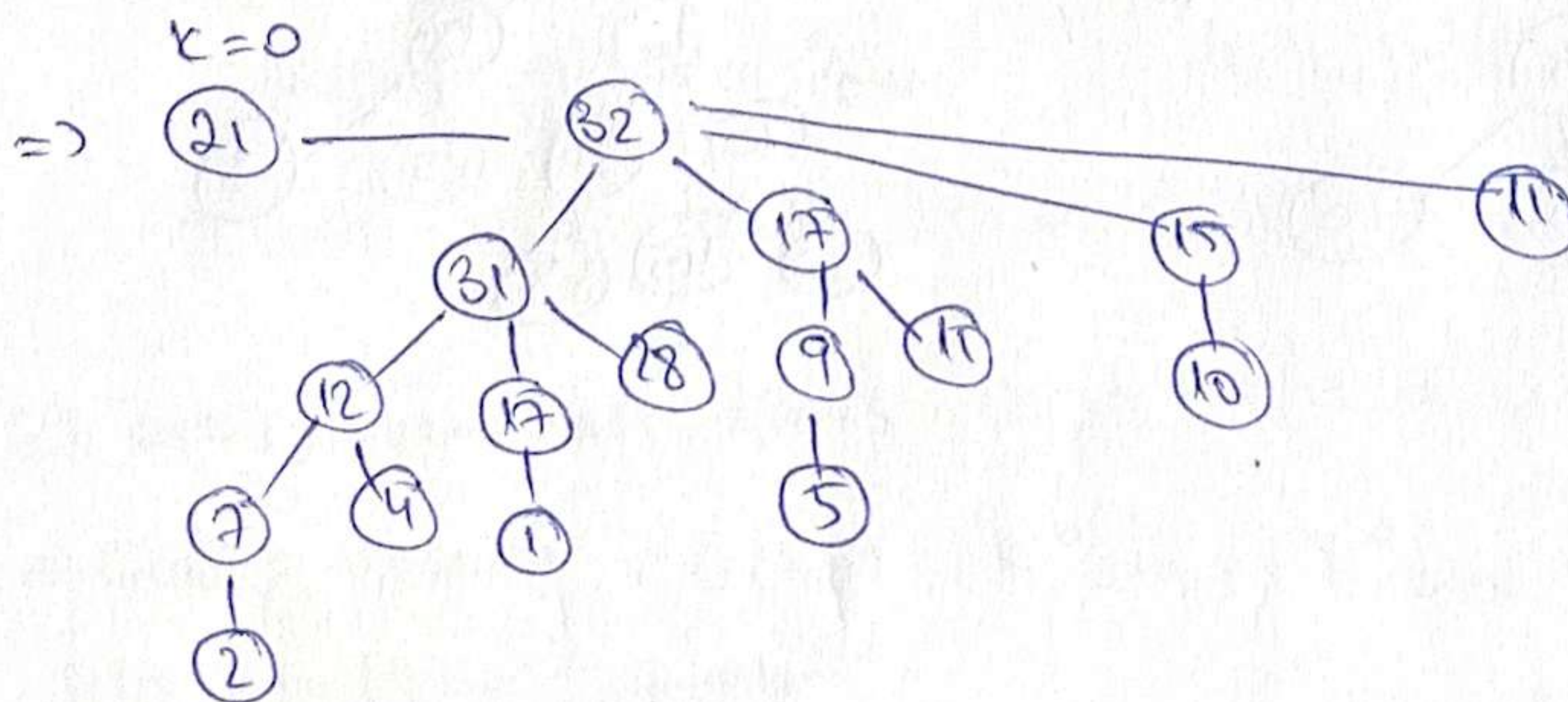
Merge $\Rightarrow$



ATTACH NEW ONE TO THE LEFT!
redo.

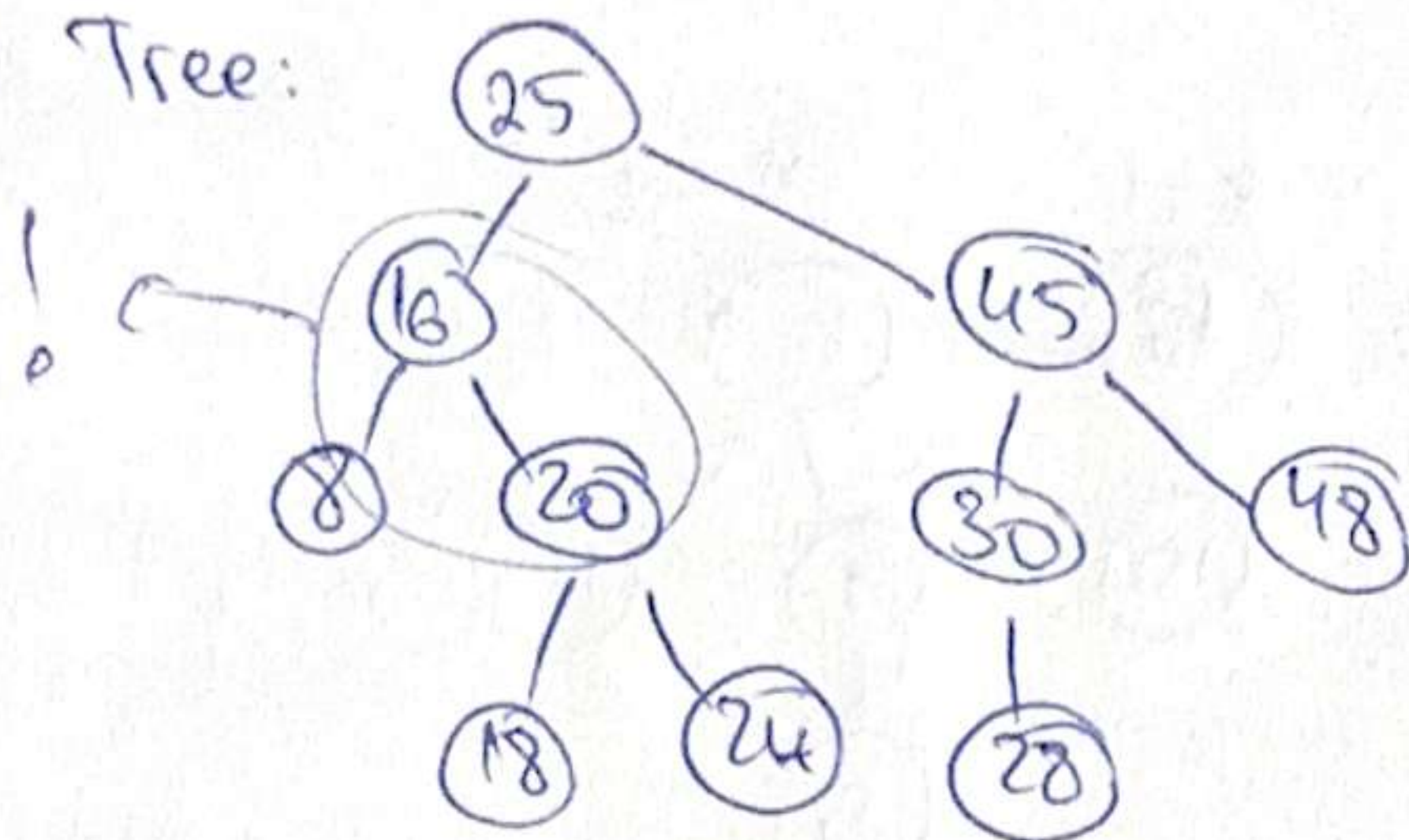


Final should be:



Remove 21 $\Rightarrow$ one single, binomial tree. Same would happen if we remove 32 (after merging)

8) Tree:



a) $\rightarrow$ AVL? b) $\rightarrow$ BST? c) $\rightarrow$ Binary Max heap?

b) BST because for (+) mode, all nodes from left subtree $\leq$, all nodes from right subtree are $>$, than the current node.

c) Not a Binary max heap because:

1. no max heap property (16 < 20 does not follow)
2. no heap structure. (empty slots)

a) AVL because $\rightarrow$ BST $\checkmark$

every node has Balance Factor $\in \{-1, 0, 1\}$

$$\text{BF}(\text{node}) = \text{height}(\text{left}) - \text{height}(\text{right})$$

$$8, 18, 24, 28, 48 \rightarrow \text{BF} = 0. \checkmark$$

$$20 \rightarrow \text{BF} = 1 - 1 = 0. \checkmark$$

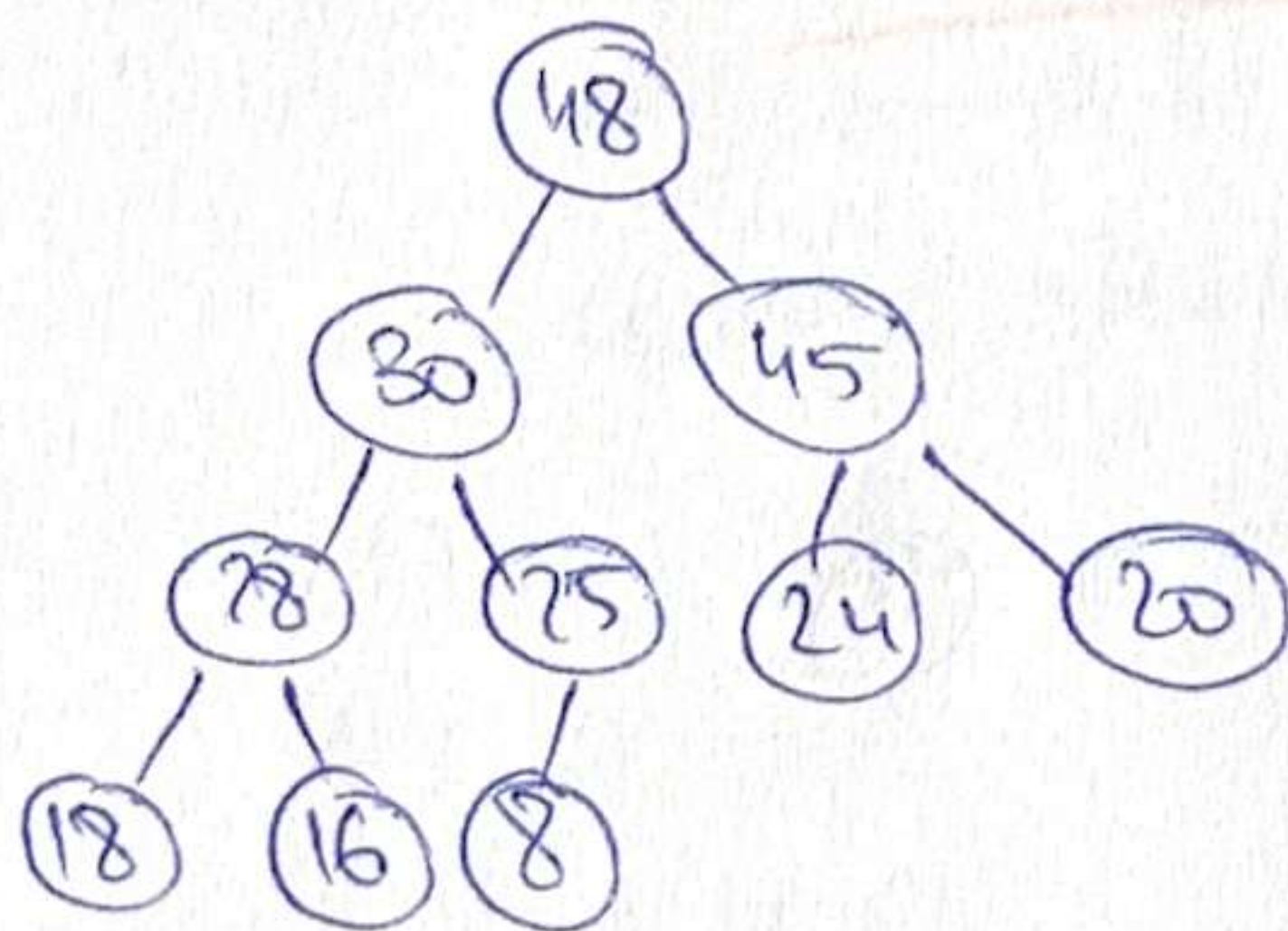
$$30 \rightarrow \text{BF} = 1 - 0 = 1 \checkmark$$

$$16 \rightarrow \text{BF} = 1 - 2 = -1 \checkmark$$

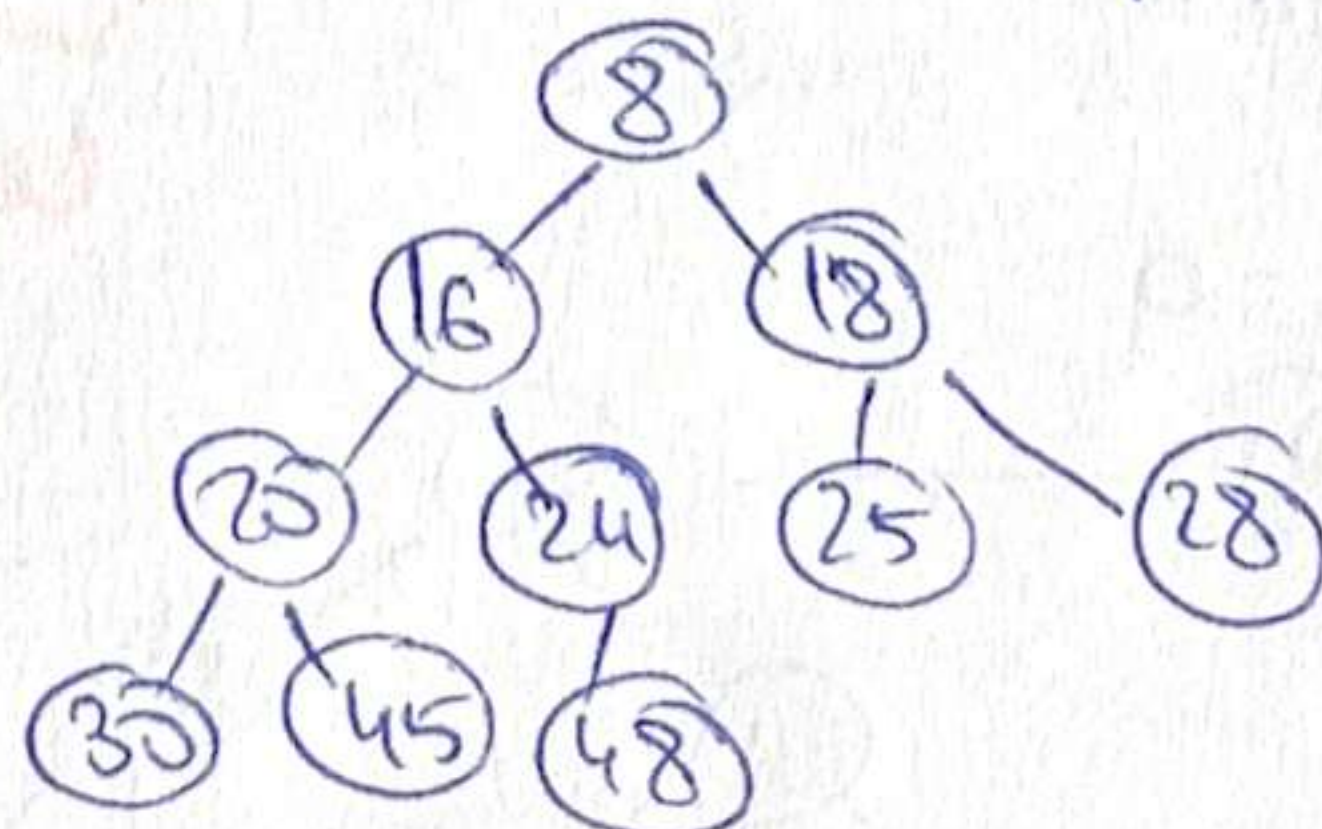
$$45 \rightarrow \text{BF} = 2 - 1 = 1 \checkmark$$

$$25 \rightarrow \text{BF} = 3 - 3 = 0 \checkmark$$

c) $\rightarrow$ make it a max heap:

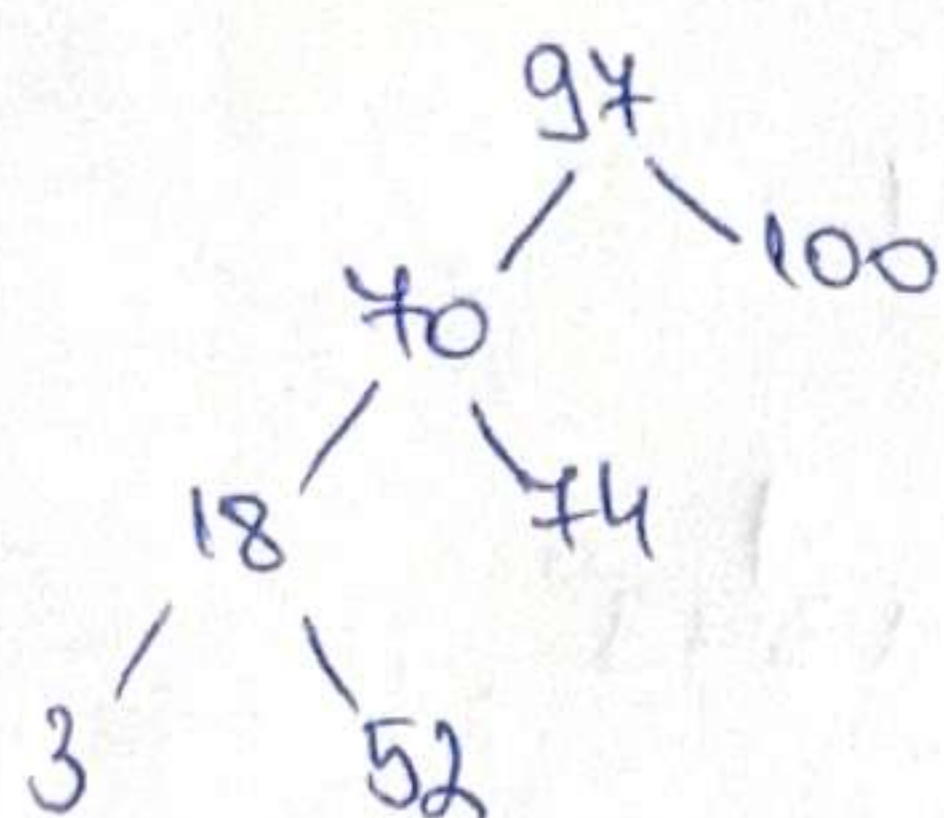


$\rightarrow$ make it a min heap:

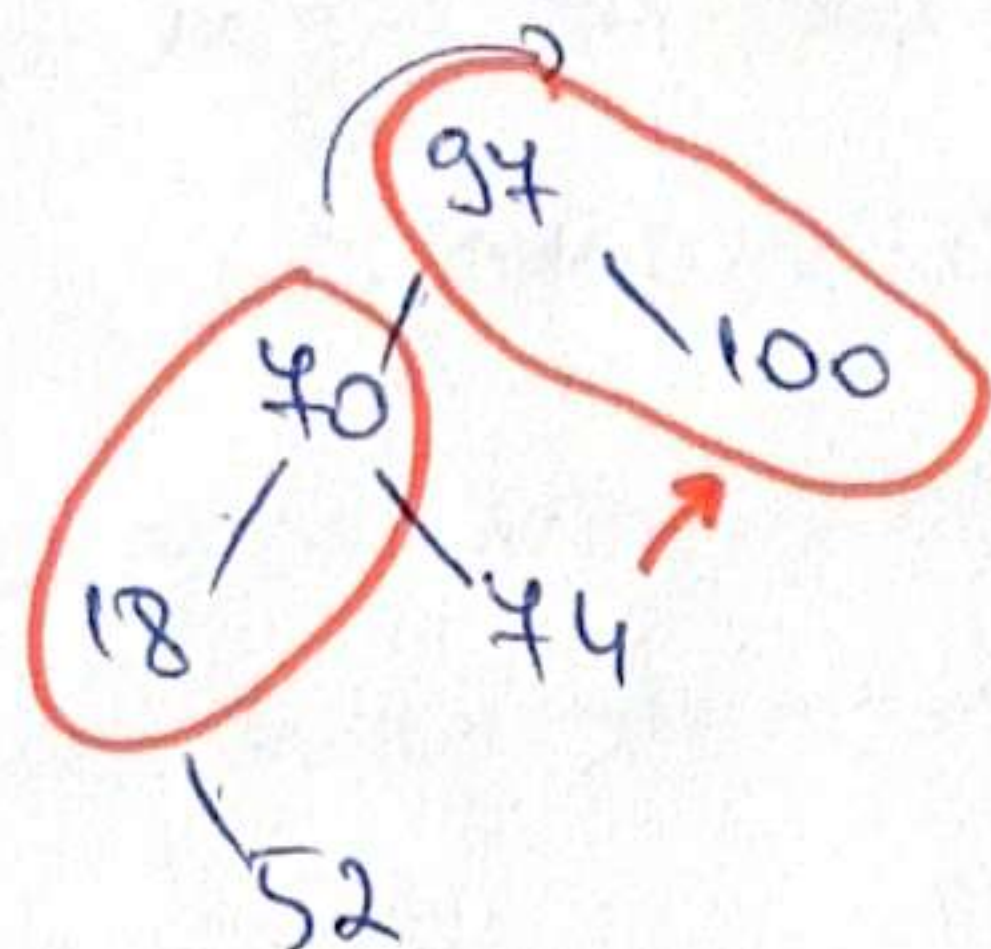


9) 97, 100, 70, 18, 74, 52, 3.

1. insert into a BST:



2. insert into an AVL:



$$BF(97) = 0 \checkmark$$

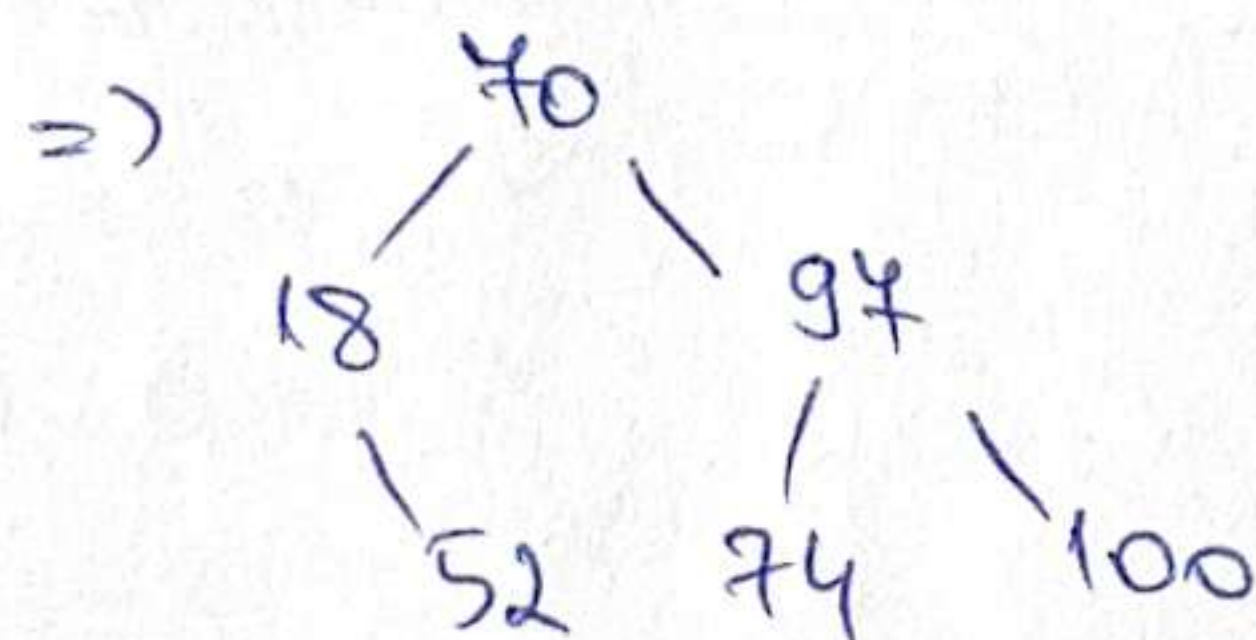
$$BF(70) = 3 - 1 = 2$$

$$BF(18) = 2 - 1 = 1$$

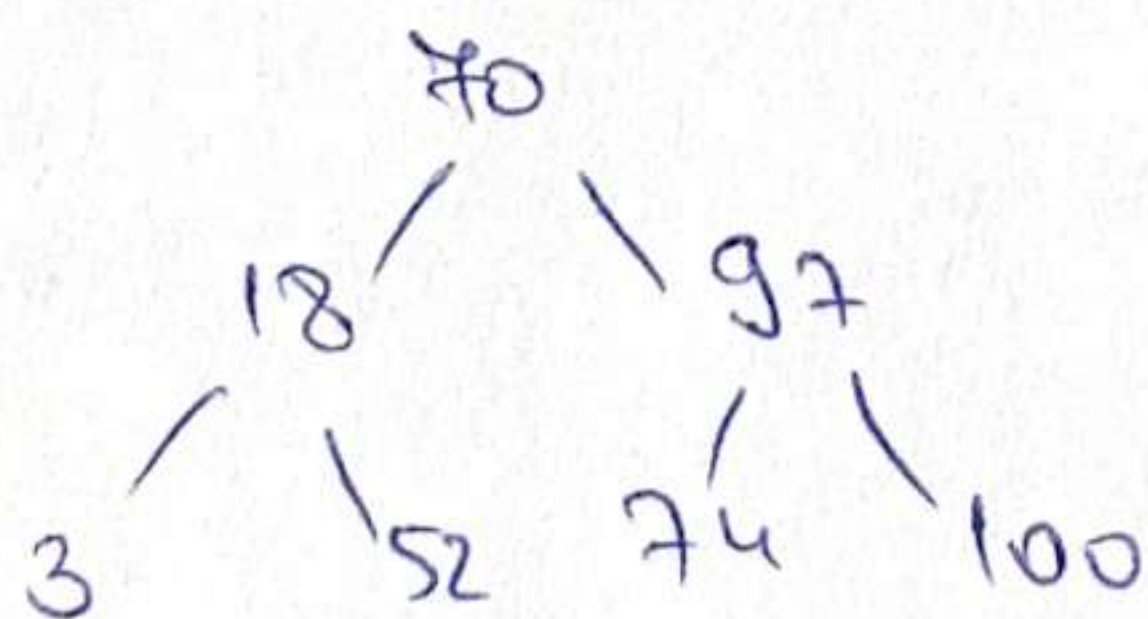
$\Rightarrow$ unbalanced (97 is the unbalanced node)

$\Rightarrow$ heavy left

LL $\Rightarrow$ single rotation to the right.



add 3: $\Rightarrow$ final:



10) Solve the recurrence: $T(n) = \begin{cases} 1, & \text{if } n \leq 1 \\ 1 + 3T(n/3), & \text{otherwise} \end{cases}$

Assume $n \gg 3$ and $n = 3^k$, $k \in \mathbb{N}$

||

Suppose $m = 3^k$

$$T(3^k) = 3T\left(\frac{3^k}{3}\right) + 1 = 3T(3^{k-1}) + 1$$

$$T(3^{k-1}) = 3T(3^{k-2}) + 1$$

$\vdots$

$$T(3^1) = 3T(3^0) + 1$$

(*)

$$\Rightarrow T(3^k) = 3T(3^{k-1}) + 1$$

$$= 3(3T(3^{k-2}) + 1) + 1$$

$$= 3(3T(3^{k-3}) + 1) + 1 + 1$$

$\vdots$

$$= 3^k \cdot \underbrace{T(3^0)}_{T(1)=1} + 3^{(k-1)} + 3^{k-2} + \dots + 3 + 3^0$$

$$= 3^k + 3^{k-1} + 3^{k-2} + \dots + 3^1 + 3^0$$

$$\Rightarrow T(3^k) = (3^{k+1} - 1) / 2$$

$$\Rightarrow T(3^k) \in O(3^k)$$

$$n = 3^k$$

$$\Rightarrow T(n) \in \Theta(n)$$